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a) Governments may intervene in markets to support firms by given them a subsidy. A subsidy is a payment or grant given to a producer to support the firm by lowering its costs of production. This in turn encourages the firm to supply more of that good. Goverments may also try to promote equity in a market, which can be done using a maximum price. (Equity means fairness) A maximum price is imposed by the government and prevents producers from the raising prices for goods above it. Oftentimes this is used to ~~promote~~ prevent monopolies from ~~the~~ controlling price, and to increase the affordability of a necessity good.

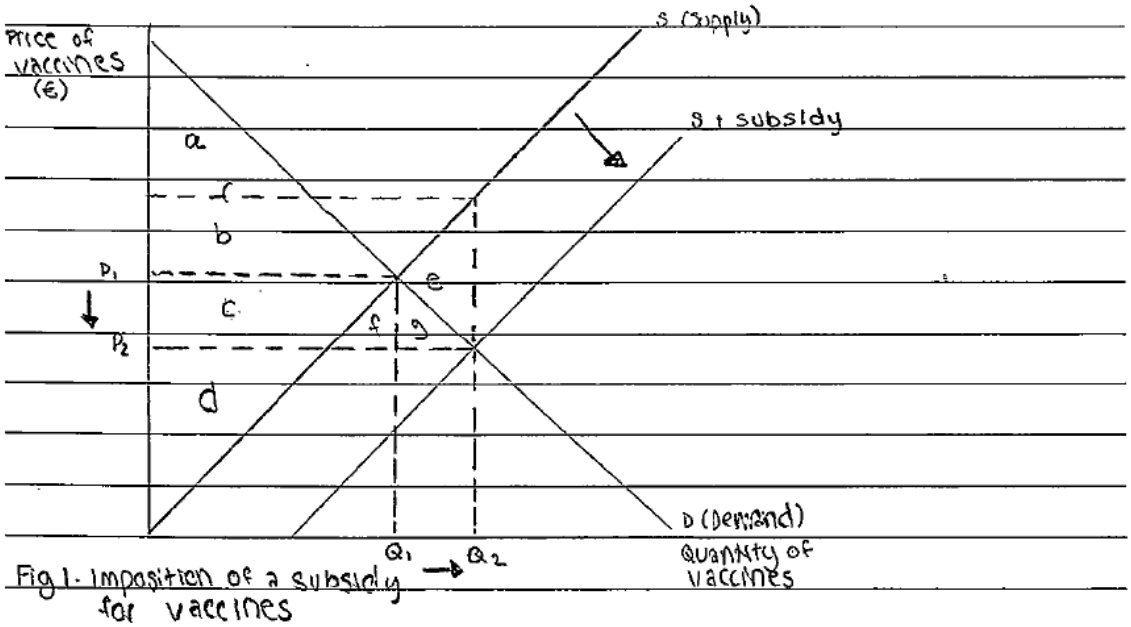


Fig 1. Imposition of a subsidy for vaccines

When a government ~~imposes~~ grants a subsidy to, for example a company producing vaccines, the supply curve shifts down from S_1 to S_1 + subsidy. The reason this is, is because ~~this gover~~ money reduces firms costs of production, as they can use the government money. Thus by lowering costs of production, firms are capable of producing more of that good. The increase in the supply of the good allows the price of vaccines (as seen in figure 1), to drop from P_1 to P_2. Thus producers are selling their good at a lower price, which encourages consumers to buy more of it. This is seen in the ~~shift~~ increase in quantity demanded of vaccines rising from Q_1 to Q_2. Both consumers and firms benefit from this policy, but the policy is mainly geared towards helping firms. However, as seen in Figure 1, consumer surplus does increase from a+b to a + b + c + f + g. Increased spending by consumers also increases producers profit.

On the other hand, governments also intervene to promote equity. While there are many ways of doing this, maximum prices are there to promote equity for consumers and producers.

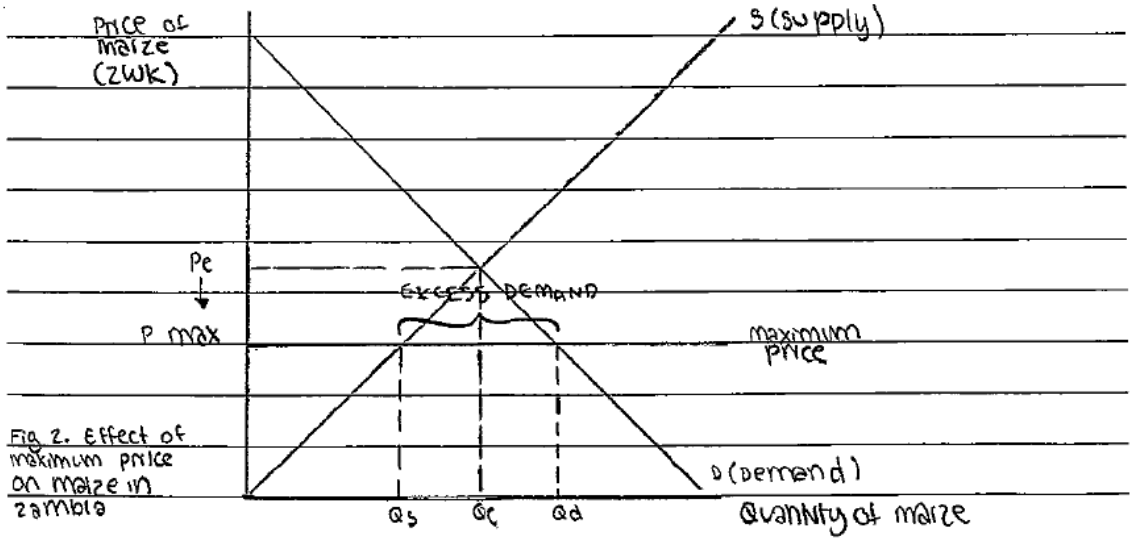


Fig 2. Effect of maximum price on maize in zambia

maximum prices reduce the price producers are able to charge for a product to below the equilibrium Pe. The maximum price in figure 2 is Pmax. Thus producers can only sell their good, such as ~~maize~~ maize, at or below Pmax. This is important particularly with necessity goods. (*) Governments impose maximum prices to ensure that the good becomes more affordable to more people. This allows for more equity to exist, as the distribution of the good is more fair. As seen in the diagram, at the maximum price more consumers are able to afford the goods, seen with quantity demanded increasing from Qe to Qd. The problem with this, however, is that it creates a situation of excess demand, as ~~the~~ quantity supplied (Qs on the diagram), ~~does~~ does not meet Qd. This results in long queues, rationing and potentially black markets developing.

~~However, in cases~~

(*) In Zambia, maize is a staple part of peoples diet, and to make sure people are nourished and everyone can purchase the good maximum prices were imposed.

Maximum prices may also make the market more equitable by reducing the power monopoly firms may have on the price. This makes smaller firms more competitive.

In conclusion, subsidies encourage the production of certain products by supplying producers who now have lower costs of production. Maximum prices are also imposed by governments to make particular goods more affordable and reduce the power monopolies may have on ~~price~~ determining the price a good can be sold at.

b) An indirect tax is a tax on expenditure placed by the government on certain goods. These are often imposed to discourage the consumption of a good, often due to negative externalities, or to earn the government revenue. Stakeholders are people who "hold a stake" or who are affected by some sort of economic situation. These may include consumers, producers and the government.

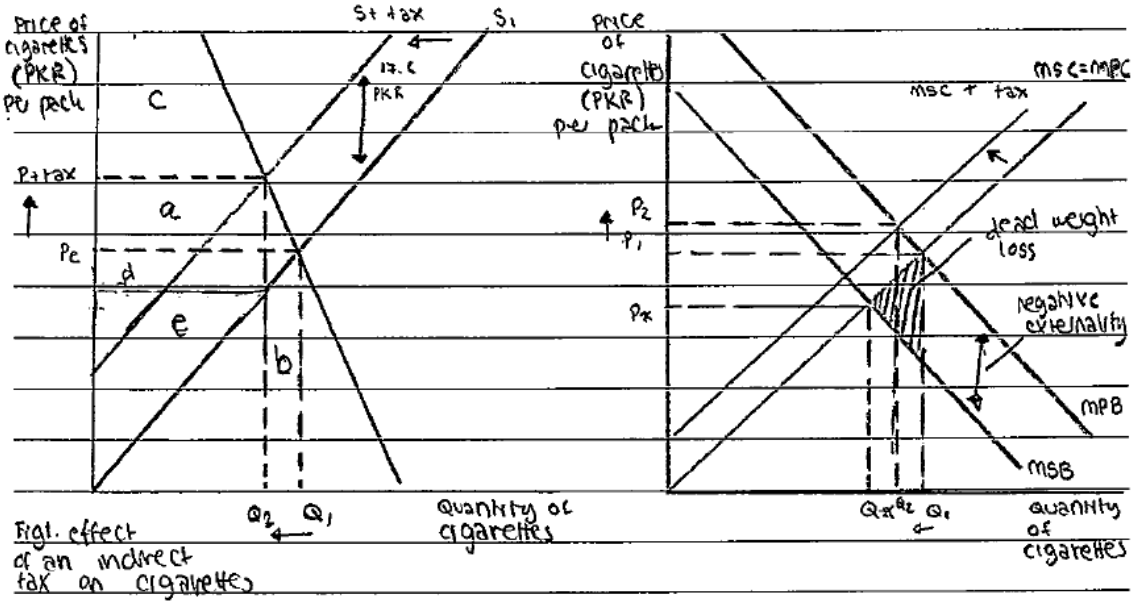


Figure 1. effect of an indirect tax on cigarettes

~~increase~~

~~Indirect taxes shift the supply curve~~ Indirect taxes shift the supply curve from S_1 to S+tax, as ~~the~~ taxes are a non-price determinant of supply. Thus a decrease in supply raises the price of the good from Pe to P+tax. Due to the higher price, quantity demanded of ~~their~~ the good falls from Q_1 to Q_2 as seen in figure 1. In 2013 in Pakistan an indirect tax on cigarettes of 17.6 PKR per pack was imposed. The idea behind this was to reduce the consumption of this demerit good, which not only had negative effects on the consumers themselves, such as health issues ~~and decreased productivity in the workforce~~ which potentially lead to death and cancer.

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b) but also creates a negative externality of consumption, as smoking hurts third parties such as second-hand smokers, or creates a burden on the healthcare system among others. This is seen in Figure 2, where marginal ~~social~~ private benefit of consuming cigarettes is greater than the marginal social benefit, creating a negative externality between the ~~MSC~~ MSB and MPB lines. The deadweight loss (shaded area) is due to the loss to society ~~from~~ due to smoking such as ~~profit~~ loss of labour productivity and burden on the healthcare system. A tax, however, moves marginal social cost up, and thus raises the price of cigarettes from P_1 to P_2, resulting in a further decreased consumption due to quantity demanded falling from Q_1 to Q_2, as the product becomes more expensive. This brings quantity demanded closer to the equilibrium Q^*, reducing the negative externality.

~~indirect~~

The effect of an ~~a~~ tax on consumers is that as the price increases quantity demanded falls, as ~~the~~ consumers are less willing to pay higher prices for the good. This also means that their consumer surplus falls from a + c to just c (Figure 1). However, it must be noted that in the example mentioned, cigarettes are ~~in~~

price inelastic good (-0.87 PED)
^~~Inelastic good~~ for demand meaning consumers are less responsive to changes in price, explaining why the increase in price (a) is not proportionate to the fall in quantity demanded (b). Area a is greater than area b. Thus the indirect tax may not have as much of an effect on consumers. As well, it depends what type of consumer it is, as the Palmerston government noticed that the youth was more responsive to the price change, than older smokers, as they had overall a lower income. Indirect taxes are also regressive, meaning that they take up a greater proportion of low income - households' income than more wealthy consumers. Another effect of indirect taxes on consumers, is that a black market for the good may develop, ~~if it is, it's~~ particularly when the price elasticity of demand is quite inelastic. Thus the ~~effect~~ tax may result in people looking elsewhere to purchase the good.

Producers carry some of the tax burden as well, indicated as area d. However, if the good is inelastic, they carry less of a burden as consumers pay area a and producers only d. ~~As well with a higher price, producer revenue does rise from~~ Due to the tax and a fall in demand producers may lose some of their revenue, however in the case of cigarettes, an inelastic good, this may not actually be the case, as price rises by a lot, and quantity demanded doesn't fall by that much. But this is not necessarily the case for other goods.

Lastly, the effect of an indirect tax on the government is primarily the revenue they receive. This is area a and d ~~on the~~ in figure 1. This revenue can be used to increase the government budget and can potentially be used to subsidise firms or spend money on other projects. It may also be used to further ~~sub~~ reduce the negative externality created by cigarettes, for example, as many countries governments promote anti-smoking ~~ad~~ ads or stop-smoking campaigns / education centres. It can also be given to the healthcare system to relieve their burden. The government benefits greatly from indirect taxes, as seen in the Massachusetts government who earned 115 million USD from a cannabis tax in 2019.

Thus, to conclude, consumers are negatively affected by an indirect tax, as they have to pay a higher price, reducing their consumption of the good. The effects of indirect taxes on consumption, however, may vary depending on whether the good is inelastic or not. Producers lose consumers, and are hurt by this, however, in the case of inelastic goods, ~~they~~ their revenue may actually increase. Lastly, governments gain revenue from indirect taxes, which can be used for other campaigns or schemes to reduce the consumption of particularly demerit goods.

OUTLINE:

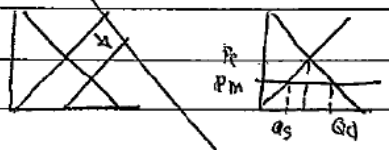
a) D: equity; fairness

subsidy

maximum price

Examples:

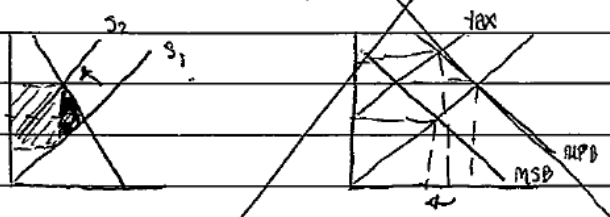
- subsidy ~~farmers~~ private producers (vaccines)
- max price zambia maize 2600 2016



b) D: indirect tax: tax on expenditure

- Stakeholders: who is affected

demerit goods: negative effect for ~~3rd par~~ people



Evaluation:

- regressive → consumers
- black market
- gov revenue
- producer burden for tax (good potentially less demanded)

* education etc. combination